



Microbiome Analysis in the Photo Sequencing Batch Reactor (PSBR) for Wastewater Treatment

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1. Introduction

Water eutrophication has emerged as a severe problem on a global scale, causing the loss of enormous amounts of resources each year¹. To solve the problem, more effective swage nitrogen-removal techniques are needed. Currently, algae-based wastewater treatment is gaining popularity because of its capacity to treat the swage without aeration, which consumes the most energy at wastewater treatment plants (WWTPs), and simultaneously produce raw biomaterials for diesel or methane². For algae-based treatment, previous research has discovered that microalgae-activated sludge system would have higher nitrogen removal effectiveness compared to pure algal system because it could form highly stable algal-bacteria granular sludge (AGBS), recycling the nitrogen at a remarkable speed^{3,4}.

The cross-feeding interaction within the bacteria and microalgae may contribute to the high efficiency of AGBS system. In this interaction, on the one hand, microalgae can provide fixed organic carbon to its surrounding bacteria and secrete various beneficial chemicals (acetate, penetrants and fatty acids, extracellular polymers, cell lysis products, etc.), supporting the growth of a variety of bacteria and providing a favored environment for them to attach. On the other hand, bacteria could supply microalgae with vitamins and auxins for growth. Cross-feeding interactions could increase the growth rate of both microalgae and bacteria as well as form extracellular polymer (EPS) to significantly improve the nitrogen recovery efficiency^{4,5}. In this research, the photobioreactor is constructed based on the common microalgae-activated sludge system. Figure.1 showed the expected biological denitrification process, which is mainly achieved by Ammonia Oxidizing Bacteria (AOB), Nitrite Oxidizing Bacteria (NOB), and Denitrifying Bacteria (DB)⁶.

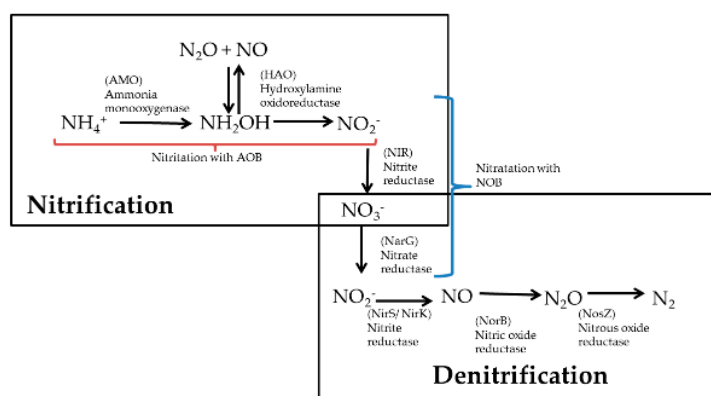


Figure.1. The processes of biological denitrification, as a technique commonly used in wastewater treatment. This biochemical reaction could convert the ammonium into nitrogen, involving nitrification (with AOB) and denitrification (with DB) steps. In particular, nitrification (with NOB) is optional.

In the preliminary investigation, it was discovered that numerous types of microorganisms had concentered into microbial communities in the photo sequencing batch reactor (PSBR) during its operation⁷⁻⁹. However, due to the scarcity of microbial community analysis in PSBR, the variance in microbiomes, functional dominating strains in different reaction phases, and interspecies connections are yet unknown. This study aims to explore and understand the development of microbiota in the operating PSBR, which could provide some possible future research directions and be beneficial in microbial selections in wastewater treatment.

2. Materials and Methods

2.1. Reactor operation and sampling

The photobioreactor with a working volume of 4 L (with an inner diameter of 7 cm and height of 45 cm) was used in this study. The inoculated activated sludge (AS) was 2.2 g/L short-cut nitrifying sludge collected from long-running lab-scale SBRs. Initial mixed liquor concentration (MLSS) of AS in the reactor was adjusted to 3 g/L before the start of the experiments. The indigenous microalgae from the raw water and inoculated sludge would gradually enrich together to make it an AGBS-dominated reactor.

External aeration was not applied during the whole experiment, and a magnetic stir was employed at 150 rpm to homogenize the liquid. A LED light was twined around the surface of the reactor with a light intensity of 6000 lux during light exposure. The reactors were maintained at room temperature and were operated in three different phases (Table.1). The reactors were fed with synthetic low C/N (1:1) ratio wastewater, with 100 ± 5 mg/L $\text{NH}_4^+\text{-N}$, 100 ± 5 mg/L COD in the form of sodium acetate, 6.5 ± 0.5 mg/L total phosphorus (TP), 60mg/L NaHCO_3 , 20mg/L MgSO_4 , 10mg/L CaCl_2 , and 1ml trace solution.

Phase	Operation Time (Day)	HRT (h)	$\text{NH}_4^+\text{-N}$ (mg/L)	COD (mg/L)	SRT (day)	Cycle (h)	Light/dark (h)	Light intensity (Lux)
I	0-12	48	100 ± 5	100 ± 5	30	24	/	0
II	13-24	24	100 ± 5	100 ± 5	30	12	6/6	6000
III	25-End	24	100 ± 5	0	30	12	10/2	6000

Table.1 Different phases for PSBR. With the increase of reactor performance, the HRT was shortened from 48h to 24 h. In phase I (Start-up phase), the 24 h cycle is consisted of the following phases: anaerobic feeding (20 min); Mixing (23h); settling (10 min) and effluent discharge (20 min; 50% of total volume). In phase II, each cycle was reduced to 12h with a 6h dark/6 h light period, HRT was thus reduced to 24h. In the phase III, the light/dark period was adjusted to 10h light/2h dark to speed up the photogranulation process. A solid retention time (SRT) of 30 days was kept by decanting 100ml mixed suspension from the reactor per day.

Samples of the reactor's microbiota were collected at 0, 20, 40, 60, 80, 150, 220, 300, and 400 operational days. These samples were then conducted further bioinformatic analysis. The reactor's chemical sampling was also carried out on the same days.

2.2. Chemical analysis

For the chemical measurement, the concentrations of NH_4^+ , NO_2^- , NO_3^- , and MLSS, were measured by standard methods¹⁰. PN and PS were measured by the Bradford method and the phenol-sulfuric acid method, respectively^{11,12}.

2.3. Bioinformatics analysis

For the genomic analysis, bio-samples in this research were performed the same process as in Kong's report⁷. The obtained microbial genomic information was firstly analyzed through alpha diversity calculation

(calculation of Shannon and Simpson index) and beta diversity calculation (PCoA calculation) for community structure investigation. Next, on the basis of the variation in microbial abundance of 11 microbial strains with significant abundance fluctuations, the microbiota population shifts were investigated. Then, through Redundancy (RDA) analysis, the correlations between 10 abundant microbial families and the reactor's physicochemical characteristics were determined. Lastly, the covarying network of 50 most abundant microbial species in the reactor was conducted. Reactive bacteria (AOB, NOB, and DB) were specifically studied for their interspecies relations. In this research, the alpha and beta diversity calculation, population shifts analysis, and the RDA analyzing were conducted in R software (4.3.0 version), while the microbial network construction was performed on Gephi software.

3. Results

3.1. Microbial community structure analysis through alpha and beta diversity calculation

To investigate the photobioreactor's microbial community structure and its changes against operation time, the alpha and beta diversity indexes of the microbiome from samples of different time were calculated. The results revealed the uneven development between the microbial strains, and the consistency between the microbial community variation patterns and the bioreactor stages.

In terms of alpha diversity (Fig.2.A and B), the Shannon index (around 3) were high, while the Simpson (about 1) and Pielou (less than 0.5) indices were relatively low, indicating the community's highly diversified and uneven nature. Besides, Pielou and Shannon's index showed patterns that increased from 20 to 80 days, decreased from 150 to 300 days, and then increased again from 300 days to 400 days.

For beta diversity (Fig.2.C), the PCoA calculation evaluated the differences and similarities between the samples. The results found that samples could be divided into three groups based on their resemblances, which included samples of 20, 40, 60, and 80 days as the first group, 150 and 220 days as the second group, 300 and 400 days as the third group. Additionally, the day 0 sample was considerably different to other samples, demonstrating the existence of growth selection in microbial community over time. Combined with the results of alpha and beta diversity, it could conclude that the microbiome in the reactor was highly diverse and uneven, and the growth selections may occur in different reacting phases.

3.2. Microbiome population shifts and dominant strain comparative genomic analysis

3.2.1. Microbiome population shifts analysis

To have a deeper understanding of the microflora changes in the photobioreactor, 11 strains (Several strains with significant abundance changes) were explored with their abundance variation toward the reactor's operation time (Fig.2.D). The results revealed the dominant microbes in each phase. The coexistence of AOB and microalgae phenomenon and growth advantage of several microbial strains were also discovered.

According to the results, in phase I (from 20 to 80 operation days), microbe *Desertifilum sp.*, *Vampirovibrio sp.*, and *Nitrosomonas europaea* had significant growing advantage. In phase II (from 80 to 220 days), *Thauera aminoaromatica* took advantages during the whole phase while *Pyrinomonadaceae* OLB17, *Cyclobacteriaceae*

UBA2336 took at day 150. In phase III (220 to 400 days), *Pantanalinema* sp. and *Xanthomonadaceae* SCMT01 showed relatively high activity.

For reactive bacteria, *Nitrosomonas* bacteria (AOB) was prevalent in phase I, while *Thauera* bacteria (DB) was in phase II. This was in line with the expected microbial shifts in biological denitrification process, guaranteeing the occurrence of denitrification in the reactor. Besides, *Nitrosomonas europaea* in *Nitrosomonas* genus and *Thauera aminoaromatica* in *Thauera* genus showed significant growth advantages over other species in the same genus. In phase III, *T. aminoaromatica* was also found possibly in suppression by *Pantanalinema* sp. and *Xanthomonadaceae* SCMT01. In terms of NOB, *Nitrospira* A sp001567445 demonstrated a relatively low abundance during the operation period with a slight increase from 150 days.

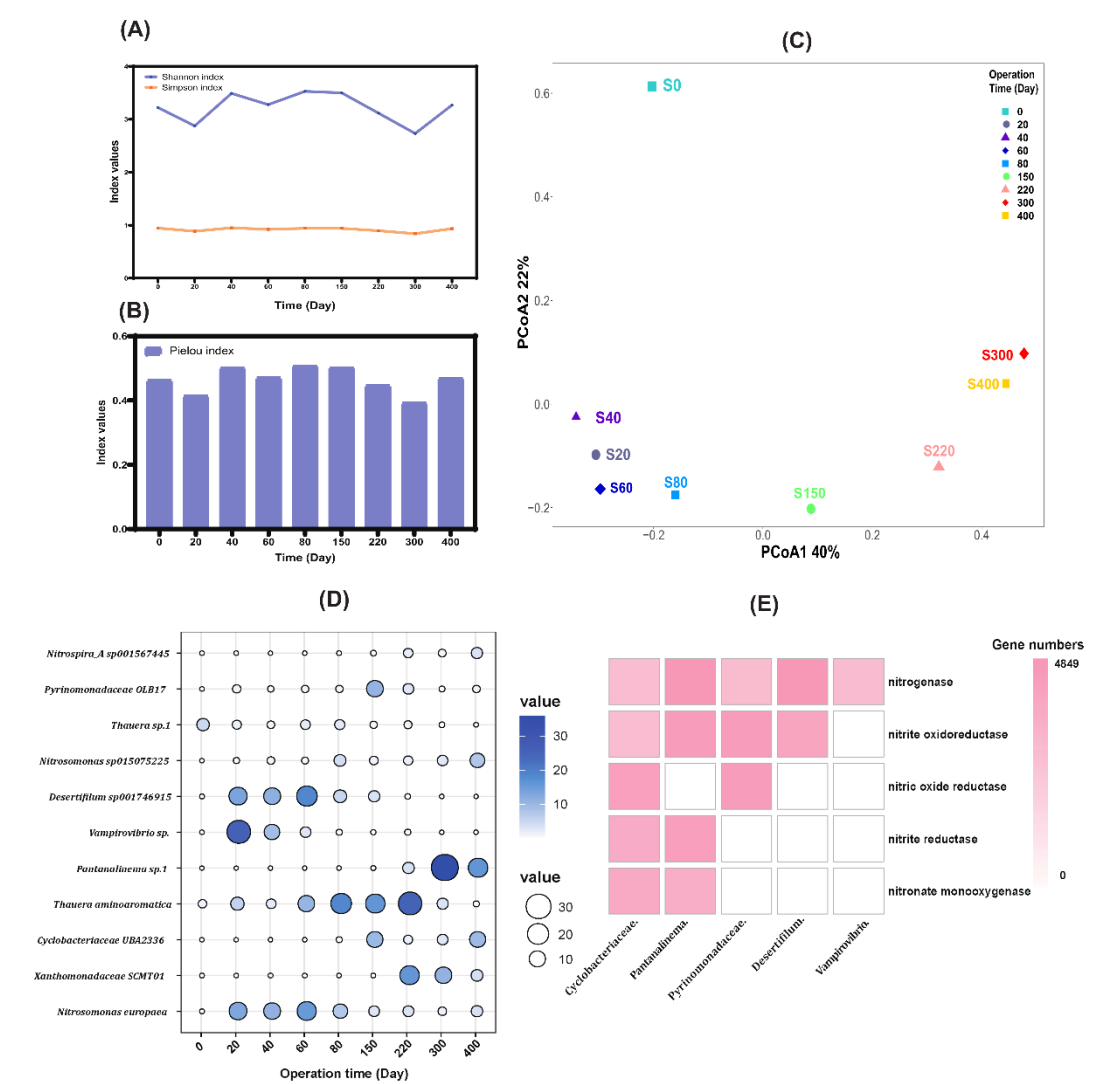


Figure.2. The results of microbial community structure, population shifts, functional gene comparison analysis. **A.** and **B.** Shannon, Simpson, and Pielou index in alpha diversity calculation for microcommunity samples in different operation time. **C.** PCoA analysis results for different operation time samples in community beta diversity calculation. The distance between the samples indicated their similarities. **D.** The bubble diagram of 11 microbial strains based on their abundance in samples from various operation time. **E.** The functional genes comparison heatmap between the non-reactive dominant strains. Microbial abbreviations from left to right were *Cyclobacteriaceae* UBA2336, *Pantanalinema* sp., *Pyrinomonadaceae* OLB17, *Desertifilum* sp., and *Vampirovibrio* sp., respectively.

3.2.2. Functional comparison of dominant non-reactive bacteria

To further understand their roles in the bioreactor process, functional comparison of non-reactive dominant strains was conducted. Presences of biological denitrification-relevant enzymes were found in these microbes excluded *Xanthomonadaceae* SCMT01 (Fig.2.E). *Desertifilum* sp. had 4849 nitrogenase and 1085 nitrite oxidoreductase genes, and *Vampirovibrio* sp. had 162 nitrogenase genes. *Pyrinomonadaceae* OLB17 had abundant genes in nitrite oxidoreductase(2532) and nitric oxide reductase(2501). *Cyclobacteriaceae* UBA2336 had all five enzyme genes, and the genes were rich in nitric oxide reductase with 1728 genes and nitronate monooxygenase with 854 genes. *Pantanalinema* sp.'s reactive genes were mainly distributed in nitrogenase (4845), nitrite oxidoreductase(2362), nitrite reductase (1934). This might suggest their involvements in synergic interactions with or competitive repression to the reactive bacteria.

3.3. Redundancy analysis of microbes towards four physicochemical parameters

To explore the relationships between the microbes and the physicochemical properties of the reactor, bacteria in 10 families were examined in their connections to four variables, including Mixed Liquor Suspended Solids (MLSS), Exopoly Saccharides (PS), Exopoly protein (PN), and Nitrogen removal rate (NRR). These correlations were listed in Figure.3. It could be found that *Rhodocyclaceae*, *Burkholderiaceae* might be critical in nitrogen removal process, having positive relations to the NRR. *Vampirovibrionaceae*, on the other hand, had negative correlations to all four parameters and was negatively associated to the denitrification reaction. Results from the RDA also shown a positive association between the NRR and extracellular polymeric substances (EPS, containing PN and PS). Therefore, EPS may be considered as another crucial component of the nitrogen removal process. Microbes from *Nitrosomonadaceae* and *Xanthomonadaceae* families were found positive correlations to EPS formation.

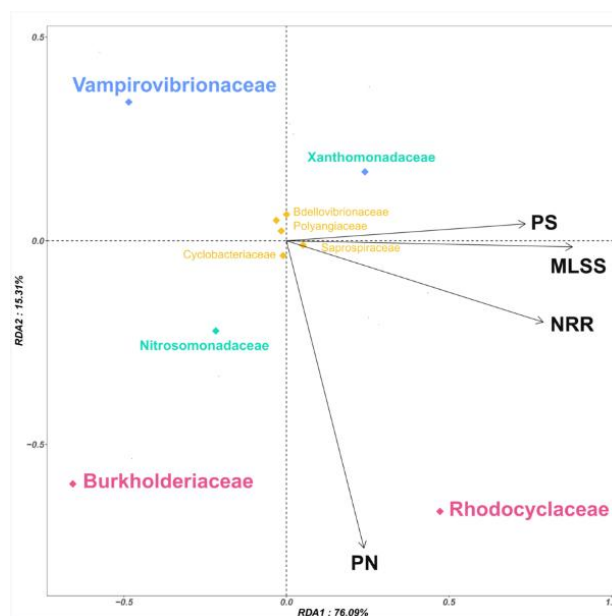


Figure.3. RDA analysis results of correlations between four parameters and 10 microbial families. Families in yellow (*Rhodobacteraceae*, *Polyangiaceae*, *Saprospiraceae*, *Cyclobacteriaceae*, and *Bdellovibrionaceae*) indicated insignificant correlations with these parameters. Families in green represented positive relations to EPS. Families depicted in pink and blue were related to NRR in favourable and negative ways, respectively.

3.4. Microbial network in the photobioreactor

The co-varying networks of 50 prevalent microbial strains were built in order to investigate the connections of various strains in the photobioreactor. Figure.4.A indicated that the microbes in the reactor had intricate relationships with each other. In the 50 microbes, reactive bacteria as *Nitrosomonas* bacteria (AOB), *Nitrospira* bacteria (NOB), and *Thauera* bacteria (DB) were found and studied their relationships to other microbes. Their relations were revealed in Figure.4.B,C,D,E.

It could be concluded that microbes in *Cyclobacteriaceae* UBA, *Saprospiraceae* and *Chitinophagaceae* would have positive interactions with AOB and NOB, while had negative interactions on DB. By contrast, *Chloroflexota* and *Gemmatimonadota* would be advent to them and favorable to DB. Microalgae groups were generally shown to have positive correlations with AOB and NOB, while have negative correlations with DB. Moreover, the preference of algae to AOB or NOB would be varied based on different species, for which was *Nodosilinea*, *Pantanalinema*, *Vampirovibrio* and *Desertifilum* favored AOB, while *Pantanalinema*, *Nodosilinea* and *Elainella* preferred NOB.

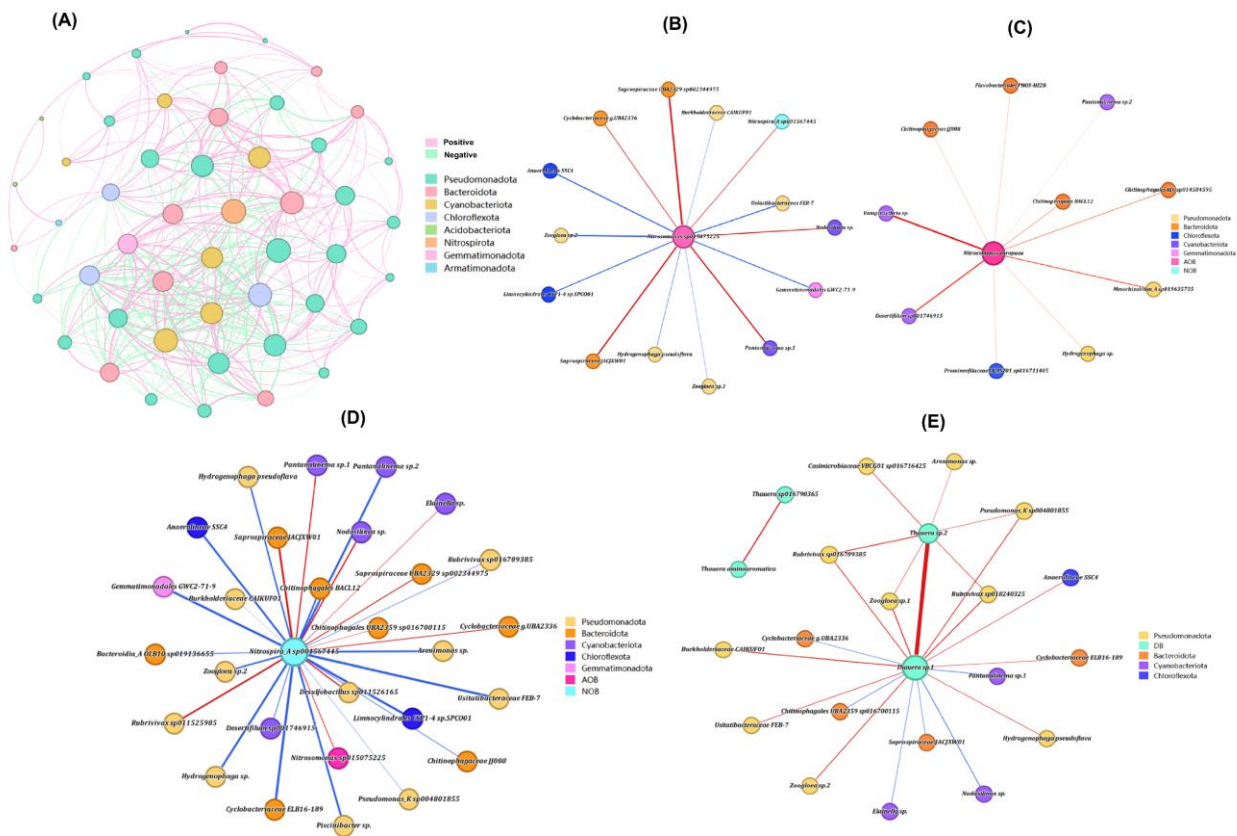


Figure.4. The microbial network in PSBR. A. The general microbial network of 50 most abundant microbes. **B. and C.** The interspecies correlations for two kinds of AOB as *Nitrosomonas* sp015075225 and *Nitrosomonas europaea*. **D.** The microbial relationships for NOB (*Nitrospira A* sp001567445). **E.** The correlations of four kinds of DB (*Thauera* sp.1, *Thauera* sp.2, *Thauera aminoaromatica*, and *Thauera* sp016790365) to other microbes. For B, C, D, E, the red color represented positive correlations while blue color represented negative relations.

4. Discussion

This research mainly studies the microbial community in PSBR without inoculation of Anammox bacteria through 4 dimensions during the operation time. Firstly, the growth selections of microbes toward different reacting phases were found in the reactor. Secondly, the construction of biological denitrification for wastewater in PSBR was proved successful through population shifts analysis.

Particularly, possible cross-feeding phenomenon of microalgae-reactive bacteria in the photobioreactor was found in this research, which was *Desertifilum* and *Vampirovibrio* microalgae with AOB. Previous studies suggested that in the cross-feeding system, the microalgae and bacteria could use each other's metabolites to support their own growth, which could increase the microbial growing speed and decrease start-up time for the reactor, improving its nitrogen-removal efficiency of the eutrophic water^{4,5,7}. This study, in addition to previous research, discovered that the possible cross-feeding microalgae and some other microbes dominating with reactive bacteria as *Desertifilum* sp., *Vampirovibrio* sp., *Pyrinomonadaceae* OLB17, *Cyclobacteriaceae* UBA2336, might also be involved in the denitrification reaction through symbiotic interaction with the reactive bacteria, producing the reaction substrates NH_4^+ from the environmental resources or reaction intermediates such as NO_2^- , NO_3^- , NO, N_2O . However, in the later stage of reactor operation (phase III, in which phase the COD input was stopped), microalgae and pseudobacteria like *Pantanalinema* sp. and *Xanthomonadaceae* SCMT01 could competitively suppress DB.

Thirdly, in the RDA analysis, microbes from *Rhodocyclaceae* and *Burkholderiaceae* families were found having positive effects on swage denitrification. These two families could have a positive impact on the denitrifying step in biological denitrification. In the report of Wang et al., *Rhodocyclales*'s important role in denitrifying process in wastewater treatment system was emphasized¹³. Similarly, *Burkholderiaceae* co-occurring with *Rhodanobacter* species have been identified as important acetate assimilators during the denitrifying process¹⁴. The situation of *Vampirovibrionaceae* was more complicated. This kind of bacteria could fix nitrogen and provide substrates for AOB in phase I (start-up period), but had general negative effects on nitrogen-removal efficiency. Possible explanation could be that *Vampirovibrionaceae* feed on the *chlorella* sp., which is an important type of microalgae for nitrogen removal¹⁵. As a result, a decline in *chlorella* sp. would cause a decrease in the reactor's nitrogen-removal capacity. Meanwhile, consistent with earlier research, this study rediscovered the positive relationships between the EPS and the reactor's denitrification effectiveness¹⁶. Moreover, from the correlations, *Nitrosomonadaceae* and *Xanthomonadaceae* could be deduced contributing in the formation of EPS.

Lastly, from the microbial network, this study discovered that the *Cyclobacteriaceae* UBA, *Saprospiraceae*, and *Chitinophagaceae* had positive correlations to both AOB and NOB, while had negative associations with DB. Besides, *Chloroflexota* and *Gemmatimonadota* were found to prefer DB and have negative connections to AOB and NOB. In terms of microalgae, *Nodosilinea*, *Pantanalinema*, *Vampirovibrio* and *Desertifilum* showed strong positive relationships with AOB, while *Pantanalinema*, *Nodosilinea* and *Elainella* had with NOB. Therefore, in the early stages of reactor operation, *Chloroflexota* and *Gemmatimonadota* should be avoid to inoculated into the activated sludge.

5. Conclusion

This study mainly investigates the microbial community in PSBR. The microbial community structure, the variation of microbiome toward the operation time, dominant microbial functions, the correlations of physicochemical parameters to microbes at family levels, and the microbial network were examined. The research finds growth selection of microbiome in different reactor stages, possible synergistic effect, and competition of some microalgae or dominant microbes to the reactive bacteria. Additionally, *Rhodocyclaceae* and *Burkholderiaceae* families' positive and *Vampirovibrionaceae*'s negative correlations to nitrogen-removal efficiency. *Nitrosomonadaceae* and *Xanthomonadaceae*'s probable involvements in EPS formation because of their positive relations. Moreover, microbial interspecies networks were interpreted. The findings of the research could offer potential directions for further research in wastewater treatment as well as suggestions in activated sludge bacteria selections. However, due to the limitations in this study, these discoveries are required for further deep-going analysis in the future.

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